



UNC
Charlotte



SMiShing: A Systematic Literature Review with Suggestions for Future Research



Brian Mellinger

Who am I?



- Undergraduate Researcher under Dr. Cori Faklaris
- Education
 - B.S. Computer Science (2027)
 - M.S. Cybersecurity (2028)
 - US Army - Cyber Warfare Officer (2029)
- My experience with cybercrime

Outline

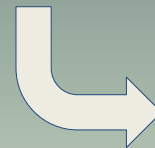
Introduction



Research



Conclusion

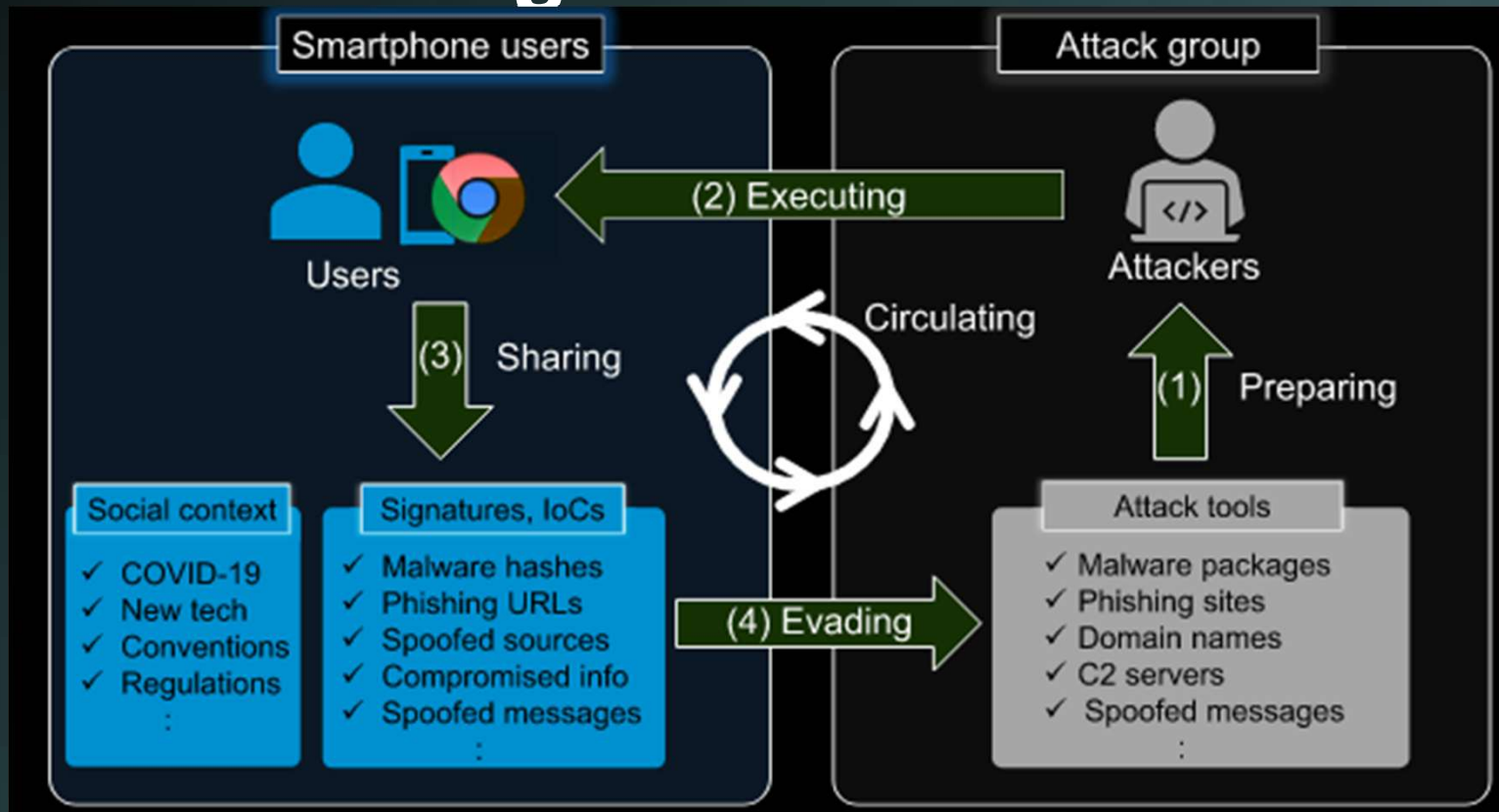


What is smishing?

“... a type of **social engineering attack** ... **fake text messages** are created ... to **obtain** user **credentials**, install malware ... click on a URL that redirects them to a phishing website, or call the provided number ...”

Md Lutfor Rahman, Daniel Timko, Hamid Wali, and Ajaya Neupane. 2023. Users Really Do Respond To Smishing. In *Proceedings of the Thirteenth ACM Conference on Data and Application Security and Privacy (CODASPY '23)*, April 24, 2023. Association for Computing Machinery, New York, NY, USA, 49–60. <https://doi.org/10.1145/3577923.3583640>

What is smishing?



Would you click it?

iMessage
Today 10:25 AM

Traffic Court Summons – Arkansas Circuit Court
Arkansas Circuit Court issues this final summons for the unpaid traffic citation AR-22834-2025.
Judge: Robert Kline
Clerk: Elena Ramirez
Case #: AR-CC-2026-TR-58374
You must appear or settle the matter in full by April 4, 2026 at 9:00 AM. Use the court portal below without delay:

<https://dfa.gov-aiude.cc/pay>

Failure to respond will lead to:

- Bench warrant issuance and potential arrest;
- Suspension of all driving privileges and vehicle registration;
- Permanent addition of violation points to your record;
- \$185 in late fees plus escalating court costs;
- Referral to collections with long-term credit and insurance impacts.

This is a binding court order.

Arkansas Circuit Court

(Reply with "Y" to open the full summons and take action.)

1. Report the violation to the DMV violation database
2. Suspend your vehicle registration starting January 19th
3. Suspend your driving privileges for 30 days
4. Charge a 35% service fee at toll booths
5. You may be prosecuted, and your credit score will be affected.

Pay Now:

<https://ncdot.gov-stwt.cc/dmv/>

Please pay immediately before enforcement begins to avoid driver's license suspension and further legal trouble.

(Reply Y to reopen this message, or copy and paste it into your browser.)

<https://ncdot.gov-stwt.cc/dmv/>

Parking Charge Notice (PCN): Our records confirm that there are outstanding parking penalty charges linked to your vehicle. The final payment deadline is 18 December 2025. Failure to remit the full amount by this date may result in the imposition of additional penalties, interest, and the initiation of further enforcement measures. In certain circumstances, continued non-payment may lead to suspension of your driving licence and may negatively impact your credit rating. To review and settle the outstanding charge, please enter your vehicle registration number using the link provided below.

<https://gov.uk-clv.bond/pay>

You are formally requested to complete payment at the earliest opportunity in order to avoid escalation to legal recovery proceedings. Thank you for your prompt attention and cooperation.

@article{timko2024smishing,
title={Smishing Dataset I: Phishing SMS Dataset from Smishtank.com},
author={Timko, Daniel and Rahman, Muhammad Lutfor},
booktitle={Proceedings of the Fourteenth ACM Conference on Data and Application Security and Privacy},
pages={289–294},
year={2024}
}

Key Takeaways

- Major issue
 - \$470 million lost in 2024 (FTC)
- Smishing is not random, its coordinated and structured
 - Professional-like in practice
- Human Psychology drives smishing success
 - Authority
 - Emotion

2025. New FTC Data Show Top Text Message Scams of 2024; Overall Losses to Text Scams Hit \$470 Million. *Federal Trade Commission*. Retrieved March 24, 2026 from <https://www.ftc.gov/news-events/news/press-releases/2025/04/new-ftc-data-show-top-text-message-scams-2024-overall-losses-text-scams-hit-470-million>

Research Questions

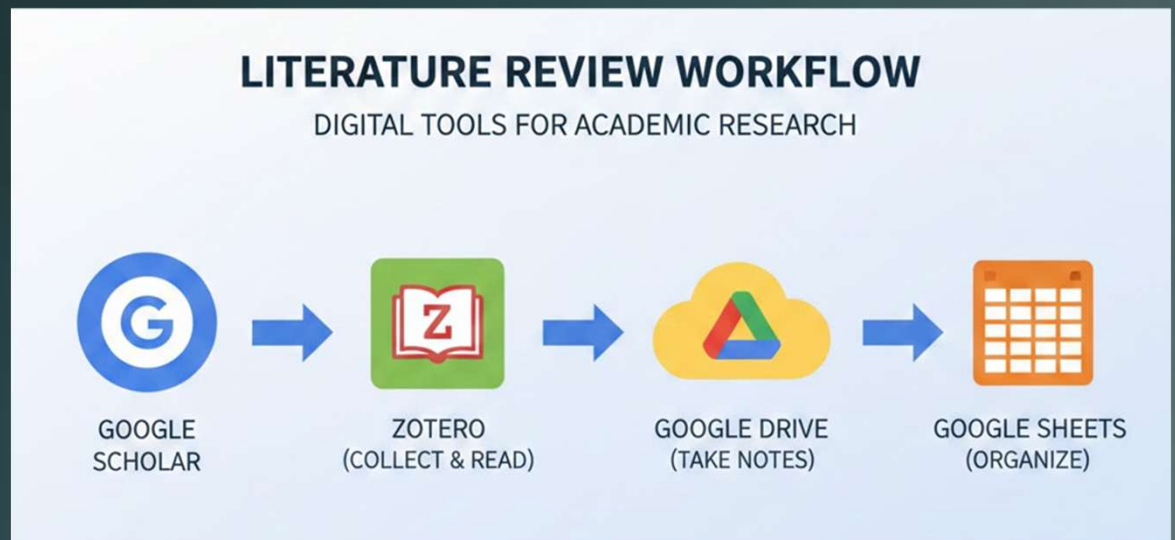
RQ1: What are the psychological and demographic factors that influence a user's susceptibility to smishing attacks?

RQ2: How effective are technical solutions in mitigating smishing?

RQ3: What gaps exist in current smishing defenses? What interdisciplinary strategies are needed to address the smishing issue?

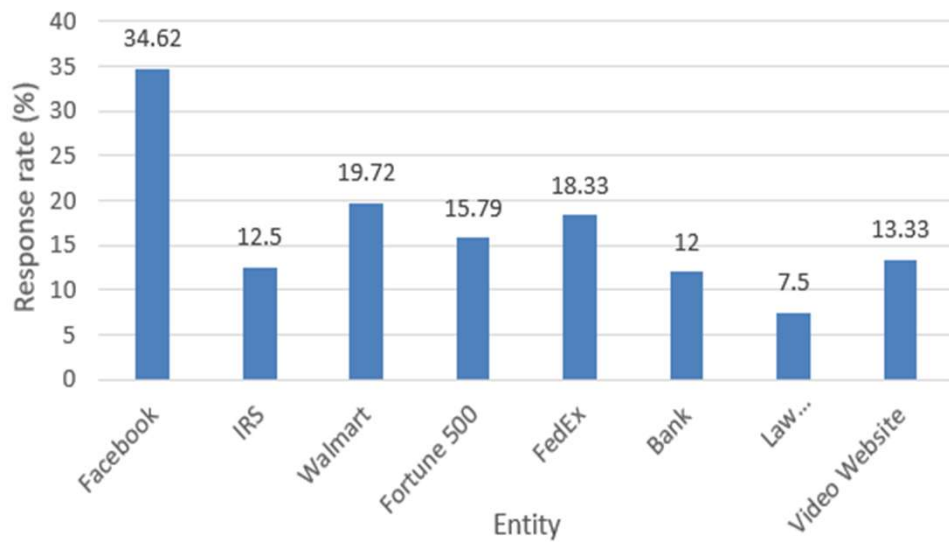
Methodology

- Systematic approach
 - Google scholar + “SMiShing”
 - Zotero
 - Google Drive
 - Sheets (22 total sources)

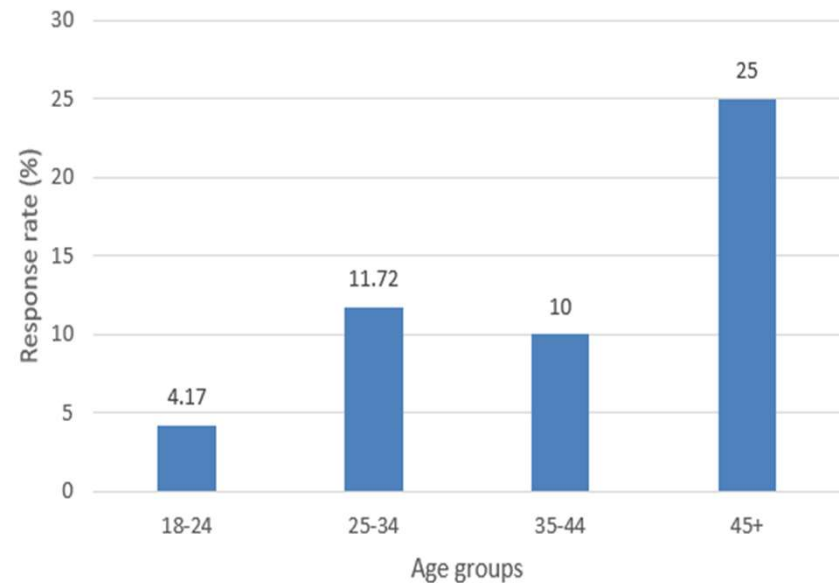


RQ1: psychological and demographic factors that influence susceptibility to smishing attacks

Response Rate (%) by Entity



Response Rate (%) Per Age Group



Charts were created by the author (Brian Mellinger)

Data from: Md Lutfur Rahman, Daniel Timko, Hamid Wali, and Ajaya Neupane. 2023. Users Really Do Respond To Smishing. In *Proceedings of the Thirteenth ACM Conference on Data and Application Security and Privacy (CODASPY '23)*, April 24, 2023. Association for Computing Machinery, New York, NY, USA, 49–60. <https://doi.org/10.1145/3577923.3583640>

RQ1: psychological and demographic factors that influence susceptibility to smishing attacks

- Authority Impersonation¹
- Fear, urgency, greed²
- African-American, Mexican-American³
- Men⁴
- Users without cybersecurity training¹

[1] Sarah Tabassum, Cori Faklaris, and Heather Richter Lipford. What Drives SMiShing Susceptibility? A U.S. Interview Study of How and Why Mobile Phone Users Judge Text Messages to be Real or Fake.

[2] Md Lutfor Rahman, Daniel Timko, Hamid Wali, and Ajaya Neupane. 2023. Users Really Do Respond To Smishing. In *Proceedings of the Thirteenth ACM Conference on Data and Application Security and Privacy (CODASPY '23)*, April 24, 2023. Association for Computing Machinery, New York, NY, USA, 49–60. <https://doi.org/10.1145/3577923.3583640>

[3] Aaron Zimba, Katongo Phiri, Chimanga Kashale, and Mwiza Phiri. 2025. Unveiling deception: a socio-economic analysis of smishing attacks on mobile money transaction users. *Humanit Soc Sci Commun* 12, 1 (December 2025), 1880. <https://doi.org/10.1057/s41599-025-06141-8>

[4] Ezer Osei Yeboah-Boateng and Priscilla Mateko Amanor. 2014. Phishing, SMiShing & Vishing: An Assessment of Threats against Mobile Devices. 5, 4 (2014).

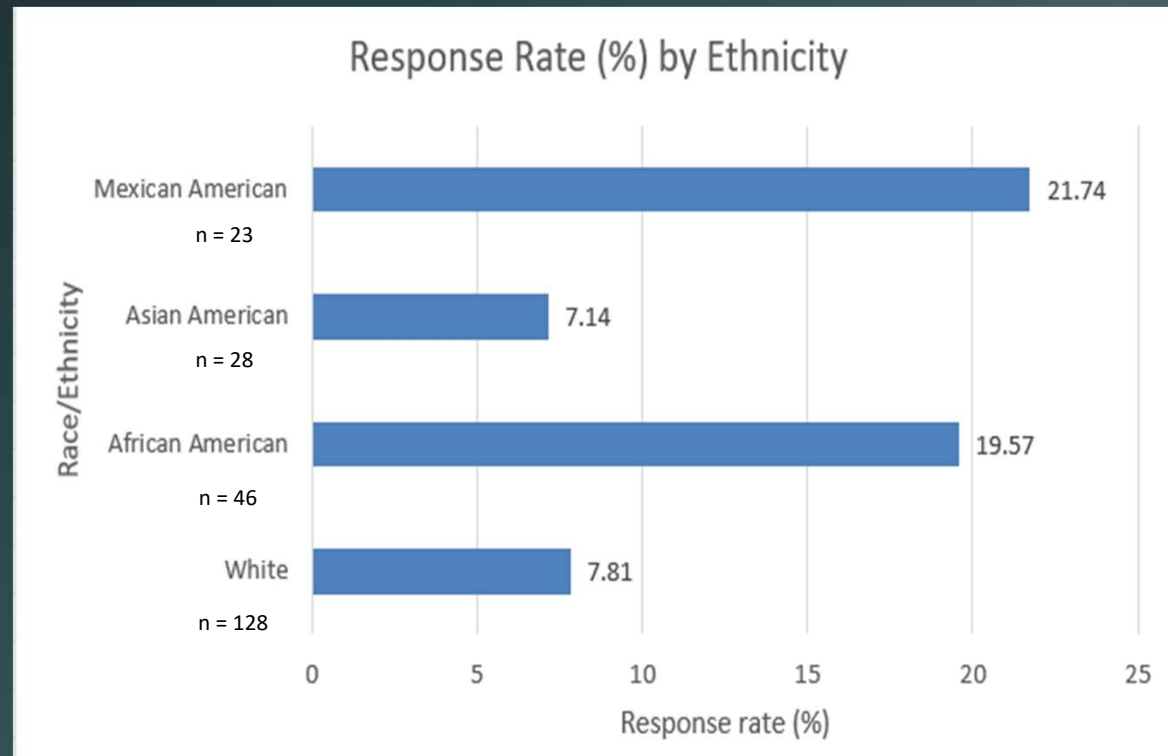


Chart created by the author (Brian Mellinger)

Data from: Md Lutfor Rahman, Daniel Timko, Hamid Wali, and Ajaya Neupane. 2023. Users Really Do Respond To Smishing. In *Proceedings of the Thirteenth ACM Conference on Data and Application Security and Privacy (CODASPY '23)*, April 24, 2023. Association for Computing Machinery, New York, NY, USA, 49–60. <https://doi.org/10.1145/3577923.3583640>

RQ2: How effective is smishing mitigation?

- Machine Learning?
 - Cybersmish (multi-analysis)¹ → 95.5%
 - SmiDCA (noun classifier)² → 96.4%
- Laws? (U.S.)
 - TCPA
 - CAN-SPAM
 - TRACED act

W. L. T. N. Kumarasiri, M. K. J. C. Siriwardhana, S. A. D. S. L. Suraweera, A. N. Senarathne, and S. M. B. Harshanath. 2023. Cybersmish: A Proactive Approach for Smishing Detection and Prevention using Machine Learning. In *2023 7th International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC)*, October 2023. 210–217. <https://doi.org/10.1109/I-SMAC58438.2023.10290228>

[1]

Gunikhan Sonowal and K S Kuppusamy. 2018. SmiDCA: An Anti-Smishing Model with Machine Learning Approach. *Comput J* 61, 8 (August 2018), 1143–1157. <https://doi.org/10.1093/comjnl/bxy039>

[2]



Image retrieved via: Seyed Mohammad Sanjari, Ashfaq Md Shibli, Maraz Mia, Maanak Gupta, and Mir Mehedi Ahsan Pritom. 2025. SmishViz: Towards A Graph-based Visualization System for Monitoring and Characterizing Ongoing Smishing Threats. In *Proceedings of the Fifteenth ACM Conference on Data and Application Security and Privacy (CODASPY '25)*, June 04, 2025. Association for Computing Machinery, New York, NY, USA, 257–268. <https://doi.org/10.1145/3714393.3726499>

RQ3: What gaps exist in current smishing defenses?

- Lack of laws specific to smishing
- The machine learning issue
- Lack of psychological research
- Ecosystem level phishing defenses
- Reporting and user support failures

RQ3: What gaps exist in current smishing defenses?

- Lack of laws specific to smishing
- The machine learning issue
- Lack of psychological research
- Ecosystem level phishing defenses
- Reporting and user support failures

7726

Discussion/Key Takeaways

- Smishing campaigns
 - organized/professional
 - Not opportunistic or rudimentary
- Current understanding
 - Human behavior is the weapon
 - Reframing the issue
- The new approach
 - Machine learning success
 - Limitations and environment testing

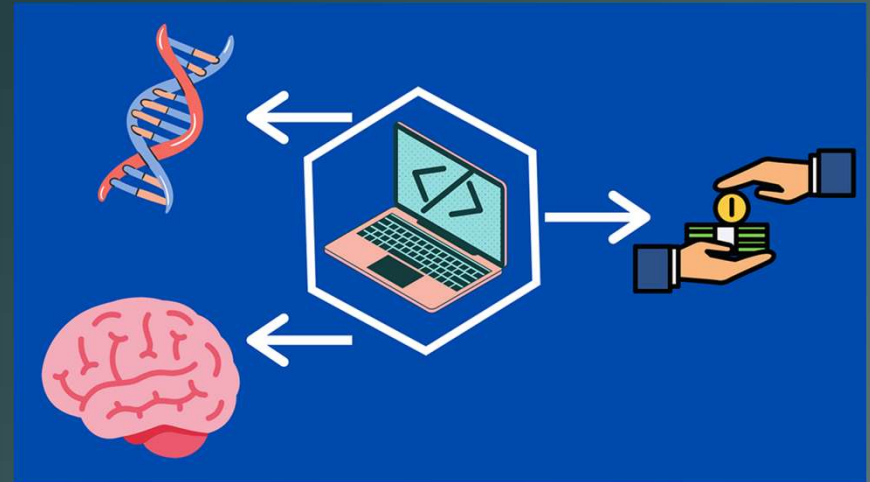


Image via: <https://afwh12.medium.com/are-computer-science-and-psychology-substantially-connected-8a7814477bea>

Future Research

- Reformed understanding of smishing
 - Psychological issue
 - mass cybercrime enterprise
- Interdisciplinary research
 - Psychology/education researchers
 - Combined research
- long standing studies
 - collection of smishing messages
 - lasting machine learning solution



Image via: <https://home.sophos.com/en-us/security-news/2021/smishing>



Conclusion

- Smishing is a \$470 million dollar issue
- Smishing is a professional business
- Smishing cannot be solved with stand alone research

Brian Mellinger
bmelling@charlotte.edu

